



KEMENTERIAN PENDIDIKAN TINGGI, SAINS, DAN TEKNOLOGI  
**POLITEKNIK NEGERI MALANG**  
**JURUSAN TEKNOLOGI INFORMASI**  
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Mata Kuliah : Basis Data Lanjut  
Program Studi : D4 – Teknik Informatika  
Semester : 3

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Jobsheet Ke- : praktikum 3

## Laporan Jobsheet

### 1. Tampilkan 5 department dengan total gaji tertinggi

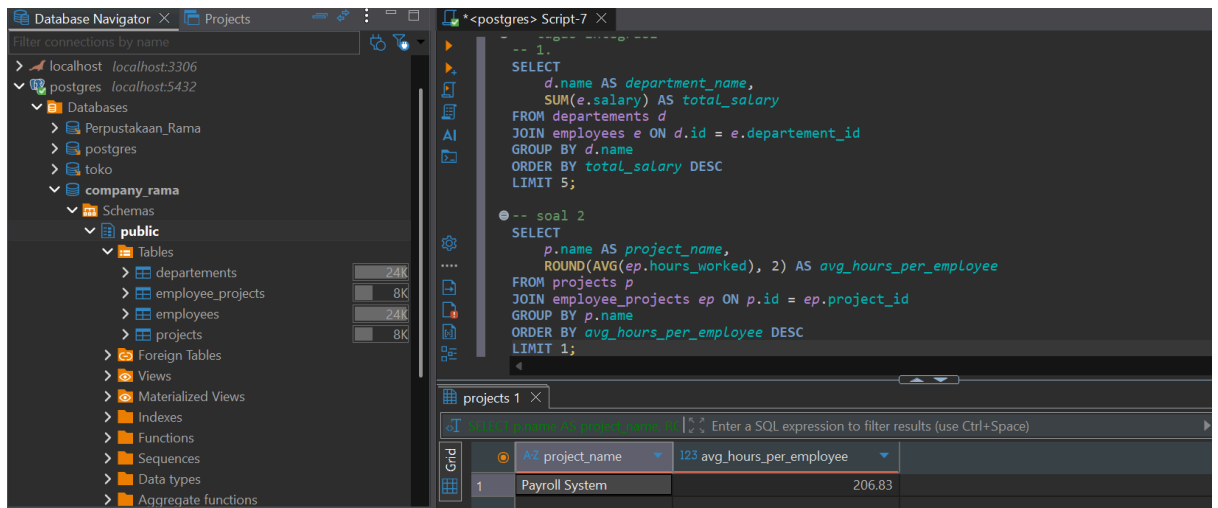
The screenshot shows the PostgreSQL interface with a query window and a results grid. The query is as follows:

```
COALESCE(ROUND(es.avg_salary, 2), 0) AS avg_salary,  
COALESCE(ps.total_projects, 0) AS total_projects,  
COALESCE(ps.total_budget, 0) AS total_budget  
FROM departments d  
LEFT JOIN dept_employee_stats es ON d.id = es.departement_id  
LEFT JOIN dept_project_stats ps ON d.id = ps.departement_id;  
  
-- tugas integrasi  
-- 1.  
SELECT  
    d.name AS department_name,  
    SUM(e.salary) AS total_salary  
FROM departments d  
JOIN employees e ON d.id = e.departement_id  
GROUP BY d.name  
ORDER BY total_salary DESC  
LIMIT 5;
```

The results grid shows the following data:

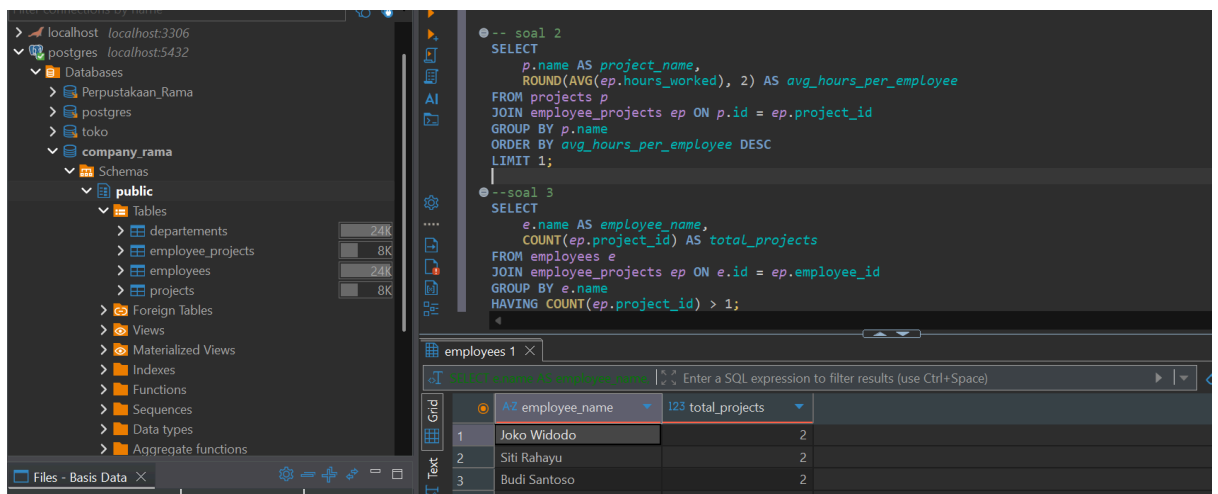
|   | department_name | total_salary |
|---|-----------------|--------------|
| 1 | Finance         | 27,000,000   |
| 2 | IT              | 24,700,000   |
| 3 | HR              | 17,500,000   |
| 4 | Marketing       | 16,000,000   |
| 5 | Operations      | 6,500,000    |

- Menggabungkan tabel departments dan employees, mengelompokkan data per departemen, menjumlahkan gaji karyawan di tiap departemen dengan SUM(), lalu mengurutkannya secara menurun untuk mengambil 5 departemen teratas.
2. Tampilkan project dengan rata-rata jam kerja per employee tertinggi 16



- Menggabungkan projects dan employee\_projects, menghitung rata-rata jam kerja per karyawan pada tiap proyek menggunakan AVG(), lalu mengurutkannya dari yang tertinggi dan mengambil 1 data teratas.

### 3. Tampilkan employee yang bekerja di lebih dari 1 project



- Mengelompokkan data berdasarkan karyawan, menghitung jumlah proyek yang ditangani masing-masing karyawan (COUNT), kemudian menyaring karyawan yang menangani lebih dari 1 proyek menggunakan klausa HAVING.
- ### 4. Buat laporan performance department: nama department, jumlah employee, jumlah project, total budget projects, rata-rata gaji, total jam kerja

The screenshot shows a PostgreSQL IDE with a SQL script in the main editor and a query result table at the bottom.

**SQL Script:**

```

WITH dept_emp AS (
    SELECT SUM(ep.hours_worked) AS total_jam_kerja
    FROM employee_projects ep
    JOIN employees e ON ep.employee_id = e.id
    WHERE e.department_id IS NOT NULL
    GROUP BY e.department_id
),
dept_proj AS (
    SELECT d.name AS nama_department,
    COALESCE(de.jumlah_employee, 0) AS jumlah_employee,
    COALESCE(dp.jumlah_project, 0) AS jumlah_project,
    COALESCE(dp.total_budget_projects, 0) AS total_budget_projects,
    COALESCE(ROUND(de.rata_gaji, 2), 0) AS rata_gaji,
    COALESCE(dh.total_jam_kerja, 0) AS total_jam_kerja
    FROM departments d
    LEFT JOIN dept_emp de ON d.id = de.department_id
    LEFT JOIN dept_proj dp ON d.id = dp.department_id
    LEFT JOIN dept_hours dh ON d.id = dh.department_id
    ORDER BY d.id;

```

**Query Result Table:**

| Grid | nama_department | jumlah_employee | jumlah_project | total_budget_projects | rata_gaji    | total_jam_kerja |
|------|-----------------|-----------------|----------------|-----------------------|--------------|-----------------|
| 1    | IT              | 3               | 3              | 160,000,000           | 8,233,333.33 | 676.5           |
| 2    | HR              | 2               | 1              | 30,000,000            | 8,750,000    | 380.5           |
| 3    | Finance         | 3               | 1              | 45,000,000            | 9,000,000    | 620.5           |
| 4    | Marketing       | 2               | 1              | 60,000,000            | 8,000,000    | 335.5           |
| 5    | Operations      | 1               | 1              | 55,000,000            | 6,500,000    | 180             |

Menggunakan **Multiple CTEs** untuk memisahkan logika agregasi agar tidak terjadi penggelembungan (*double-counting*) data saat melakukan JOIN antar tabel relasi *many-to-many*:

1. dept\_emp: Mengkalkulasi jumlah karyawan dan rata-rata gaji per departemen.
2. dept\_proj: Mengkalkulasi jumlah proyek dan total anggaran per departemen.
3. dept\_hours: Mengkalkulasi total jam kerja per departemen.
4. Query utama menggabungkan ketiga CTE tersebut dengan tabel master departments menggunakan LEFT JOIN dan fungsi COALESCE untuk menangani nilai NULL.